Calibrating the Decision, Not the Model: Per-Class Thresholds and Argmax Scaling for Training-Free Open-Vocabulary Segmentation of Remote Sensing Imagery

Mid-Project Review

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ఇండియన్ ఇన్స్టిట్యూట్ ఆఫ్ ఇన్ఫర్మేషన్ టెక్నాలజీ, డిజైన్ అండ్ మాన్యుఫ్యాక్చరింగ్, కర్నూలు

भारतीय सूचना प्रौद्योगिकी, अभिकल्पना एवं विनिर्माण संस्थान, कर्नूल

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Presentation Outline

1. Introduction — the task

Open-vocabulary semantic segmentation of aerial imagery.

- Input: an image *and a list of class names, as plain words*.
- Output: a class for every pixel.
- **Training-free / annotation-free**: no weights are trained. The class list is supplied at inference and can change without retraining.

Why it matters. A model trained for N fixed classes cannot answer a question outside them. Land cover mapping constantly needs new classes — `solar panel`, `flood water`, `debris` — and labelled data for them rarely exists. **The pipeline we build on.** SAM 3

emits, per class prompt: a presence score S_{pres}, a dense semantic map, and instance masks. SegEarth-OV3 fuses these and thresholds the result.

1. Introduction — the contribution in one sentence

A single confidence threshold is the wrong *shape* for a multi-class open-vocabulary pipeline. Fitting **one threshold per class**, and **one scale per class before the argmax**, on the same supervision budget the baseline already spends tuning its single τ — **with no weights trained** — recovers a real gain.

	LoveDA	Potsdam	OpenEarthMap
SegEarth-OV3 (reproduced)	47.38	57.60	44.16
+ per-class τ	+1.18	58.35	+0.16
+ argmax scaling	**+2.32**	**63.27**	—

All verified end-to-end by the official evaluator on held-out tiles, not by cache arithmetic.

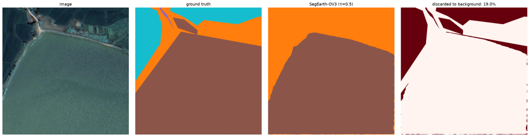
2. Motivation — the baseline throws land cover away

On LoveDA validation, at the published $\tau = 0.5$:

- **29.68%** of all real-class pixels are assigned to `background` — 323 million pixels.
- Discard outnumbers real-class confusion **3:1**. The dominant error is *silence*, not error.
- `water`: precision **89.5%**, recall **54.7%**. Right nine times in ten when it fires; finds half of what is there.

Every global fix costs more than it returns:

- $\tau \to 0.1$: -5.54 mIoU, 1.73 wrong per 1 right
- remove presence gating: -11.97 mIoU



LoveDA tile: image / ground truth / baseline / discarded pixels. Whole water regions dropped to `background`.

2. Applications — why a cheap correction matters

Disaster response. Flood extent from UAV imagery within hours. No labelled data for the affected district; no time to train.

Renewable-energy infrastructure. `solar panel` is a class no land-cover benchmark contains. Supplied as two words.

Informal settlement mapping. `tin roof`, `unpaved road`, `water tank` — vocabulary that changes per survey.

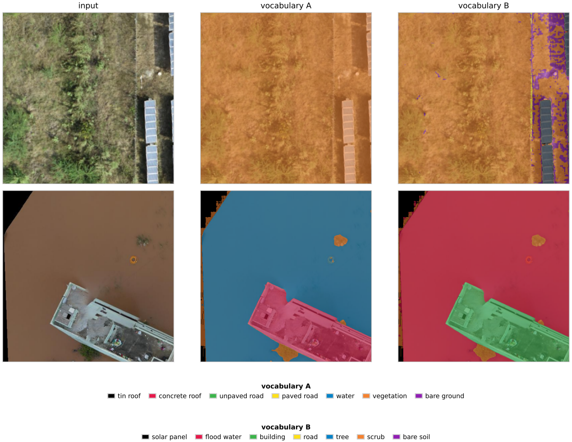
Cost of our correction: ~200 labelled tiles, minutes of CPU, **no gradient steps**. Cost of fine-tuning: thousands of tiles, a

Measured, on Indian UAV imagery

25 tiles, 5 openly licensed scenes, 4–5 cm — a region absent from all three benchmarks. Run unchanged under two class lists typed at inference.

- The partition is stable under renaming: 78.6% `water` vs 79.1% `flood water`; 18.2% `concrete roof` vs 18.3% `building`.
- The panels are only partly found: 100.0% of the solar-farm tile goes to one class when no word for them exists; 4.6% with `solar panel` supplied.

2. Applications — one scene, two vocabularies



Same weights, same forward pass; only the class list differs. **Bottom:** a flooded building — both vocabularies recover the same regions and only the names follow the words. **Top:** a photovoltaic array the settlement vocabulary has no word for. No ground truth exists for this imagery, so these are pixel shares, not accuracy.

3. Problem statement — formulation

Let $\mathcal{C} = \{1, \dots, N\}$ be the vocabulary and $s_c(p) \in [0, 1]$ the score the pipeline assigns class c at pixel p.

The baseline decision rule (one global τ):

$$\hat{y}(p) = \begin{cases} \arg\max_c s_c(p), & \text{if } \max_c s_c(p) \geq \tau \\ b \ (\textsf{catch-all}), & \text{otherwise} \end{cases}$$

What we fit instead, on a calibration set of ∼200 labelled tiles:

$$\hat{y}(p) = \begin{cases} c^\star, & \text{if } s_{c^\star}(p) \geq \boldsymbol{\tau_{c^\star}} \\ b, & \text{otherwise} \end{cases} \qquad c^\star = \arg\max_c \ \boldsymbol{w_c}\, s_c(p)$$

Objective

$(\boldsymbol{w}, \boldsymbol{\tau}) \ = \ \arg\max \ \mathrm{mIoU}_{\mathrm{real}}$ by coordinate ascent. **No network weights are touched.** The threshold reads the *raw* score $s_{c^\star}$, so $\boldsymbol{w}$ acts only on the `argmax`.

3. Why per-class τ is the *complete* family — and w is not

Claim. Given a fixed `argmax`, any strictly increasing per-class recalibration g_c composed with a global τ collapses to a per-class threshold vector:

$$g_c(s) \ \geq \ \tau \quad \Longleftrightarrow \quad s \ \geq \ g_c^{-1}(\tau) \ = \ \tau_c.$$

- So **per-class τ exhausts everything reachable by rescaling scores after the argmax.** Nothing in that family can beat it.
- Scaling *before* the argmax changes *which* class wins, so it is a **strictly larger family** — it reaches the errors a threshold provably cannot.

Measured consequence (LoveDA, §7.7)

6.0% of catch-all assignments are cases where the catch-all *won the argmax* at $s \geq \tau$. Lowering a threshold cannot change an argmax, so that residual is unreachable by lever one and is exactly what lever two addresses.

4. Literature survey — the lineage (1/2)

Work	Venue	Contribution / relevance
CLIP [1]	ICML 2021	Image–text contrastive pretraining; the origin of the open-vocabulary paradigm.
SAM [2]	ICCV 2023	Promptable class-agnostic segmentation; masks without a fixed label set.
SAM 3 [3]	2025	Presence + semantic + instance heads in one model. **Our backbone.**
MaskCLIP [4]	ECCV 2022	First to extract dense predictions from CLIP by removing pooling.
SCLIP [5]	2023	Correlative self-attention; the codebase SegEarth-OV3 derives from.
ClearCLIP [6]	ECCV 2024	Isolates which CLIP components harm dense prediction.
ProxyCLIP [7]	ECCV 2024	Uses self-supervised features (DINO) as structural guidance.
GroupViT [8]	CVPR 2022	Learns segment grouping from text supervision alone.

4. Literature survey — direct competitors and tools (2/2)

Work	Venue	Contribution / relevance
SegEarth-OV [9]	CVPR 2025	Feature upsampler + global bias subtraction for RS. Defines the protocol.
SegEarth-OV3 [10]	2025	SAM 3 for RS OVSS: dual-head fusion + presence gating. **Our baseline, reproduced at 47.38 vs 47.4.**
ConInfer [11]	2026	GMM over DINOv3 patches fused with the VLM prior; +2.80 mIoU. **Nearest competitor — our method *composes* with it.**
OVRSISBench [12]	2026	Unified open-vocabulary RS evaluation protocol.
DenseCRF [13]	NIPS 2011	The classical "use spatial context" answer we position against.
CRF-as-RNN [14]	ICCV 2015	Learned CRF inference; same family.
Calibration [15]	ICML 2017	Modern networks are systematically miscalibrated. **The premise we exploit.**
Otsu [16]	IEEE TSMC 1979	Classical automatic thresholding; one of our label-free baselines.
SLIC [17]	IEEE TPAMI 2012	Superpixels; our region atoms.
LoveDA [18] / OEM [19]	NeurIPS'21 / WACV'23	The benchmarks.

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works; full bibliography (26 entries) in the paper.

5. Research gaps

1. **The decision rule is unexamined.** Every method in this family ends in `argmax` + one global τ, tuned per dataset. All effort goes into the *scores*; nobody asks whether the *rule* on top of them is the right shape.
2. **The discarded residual is never quantified.** SegEarth-OV3 reports mIoU; it does not report that **29.68%** of real-class pixels on LoveDA are assigned to `background`.
3. **Catch-all classes distort the metric, in both directions.** mIoU is an *unweighted* mean, so a catch-all owns $1/N$ of it however meaningless. We measure +22.67 inflating on one dataset and −3.51 deflating on another.
4. **Context is modelled visually, not semantically.** ConInfer uses a per-scene visual prior over patches; nobody tests a corpus-level *semantic* class-adjacency prior. (We did — and it fails. Reported as a negative with an oracle bound.)

6. Objectives

1. **Reproduce** SegEarth-OV3 exactly, so every later number is measured against a baseline that runs on our hardware. ✓ 47.38 vs published 47.4
2. **Quantify** the residual the baseline discards, and test whether any global knob reaches it. ✓ 29.68%; all three knobs cost more than they return
3. **Explain** what governs the residual across datasets, by intervention rather than correlation. ✓ vocabulary intervention, arity-controlled, pre-registered
4. **Design** a correction that trains no weights and spends the same supervision budget the baseline already spends. ✓ two levers, +2.32 LoveDA / +5.67 Potsdam
5. **Bound** what is achievable without labels at all. ✓ 12 attempts, all bounded, with a stated mechanism
6. **Demonstrate** on imagery and a vocabulary no benchmark covers. ✓ Indian UAV scenes, 4–5 cm

7. Experimental setup

Datasets

- **LoveDA** — 1669 val tiles, 1024^2, 0.3 m, 7 classes, urban/rural split
- **OpenEarthMap** — 384 tiles, 0.25–0.5 m, 9 classes
- **ISPRS Potsdam** — 2016 tiles, 5 cm, 6 classes
- **OpenAerialMap (India)** — 25 tiles, 4–5 cm, no ground truth

Baselines

- SegEarth-OV3 (SAM 3) — reproduced
- ConInfer (CLIP + DINOv3) — run in a separate environment
- τ sweep, presence removal, per-class Otsu, 12 label-free rules

Protocol

- 5-fold cross-validation; calibration and evaluation tiles **always disjoint**
- Baseline recomputed on the *same* held-out tiles
- Both metrics reported: full mIoU *and* catch-all-excluded

Hardware

- RTX 2000 Ada, 16 GB, 70 W cap
- One LoveDA pass ≈ 24 min
- **Fitting the correction: minutes of CPU**

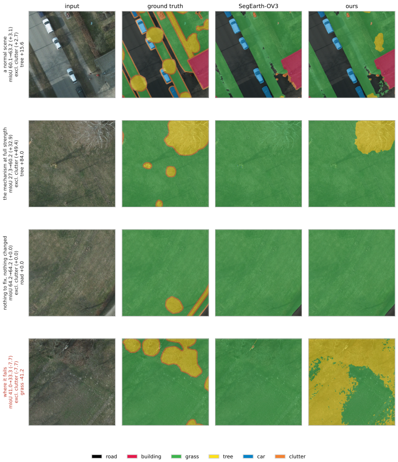
Validation gate: every instrumented run must still reproduce 47.37 mIoU and 29.68% discard, or the patch changed behaviour.

7. Results — both levers, five-fold

	LoveDA	**Potsdam**	**OEM**	**ConInfer/LoveDA**
Lever 1: per-class τ	$+1.18 \pm 0.45$	$+0.59$	$+0.30$	$+2.51 \pm 0.34$
Lever 2: argmax scaling	$+1.16 \pm 0.19$	$+4.86 \pm 0.35$	—	-0.10
Total	$+2.32$	$+5.67$	$+0.30$	$+2.51$
Catch-all-excluded	$+1.36$	$+6.00$	$+1.75$	$+1.94$
Folds positive	5/5	5/5	—	5/5

- **Two architectures** (SAM 3, CLIP+GMM), **three datasets**, two operating points. **The claim is not about SAM 3.**
- Our method *composes* with ConInfer: their pipeline 36.99 → 39.52 mIoU.
- **Potsdam tree: 37.92 → 59.09 IoU** (precision 93.34%, recall 38.63%) — exactly the asymmetry the method targets.

7. Results — the method, visually



Four Potsdam tiles. Same weights, same forward pass — only the decision rule differs. Row four is a tile where the method LOSES 7.72 mIoU and is included deliberately: the five-fold gain is +4.86±0.35, an *average*.

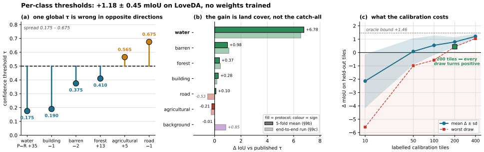
7. Results — what cannot be done without labels

12 label-free rules tested. All fail.

- Otsu, confidence percentile, presence-scaled, 8 per-class statistics, reachability
- Best: +0.42 against a random control's p95 of +0.58
- Oracle bound: +1.24

The mechanism, not just the failure:

Precision *is* predictable without labels ($\rho = +0.943$, $p = 0.017$) — and it does not determine the threshold ($\rho = -0.429$, $p = 0.419$). The optimal τ solves a **coupled multi-class IoU objective**: raising one class's threshold moves pixels into the catch-all and changes every other class's optimum. **No per-class scalar can express that.**



Fitted thresholds span 0.175–0.675 against a single global 0.5 — one value is wrong for different classes *in opposite directions*.

7. Honest limitations

- Thresholds do not transfer across a domain shift. Fitted on LoveDA-rural, applied to urban: −1.11 mIoU — *worse than not calibrating.* Pooling both domains also loses most of the gain.
- The gain is not uniform. LoveDA rural +2.77, urban +0.10. Our headline is a benchmark-level figure, not a claim about every stratum.
- No label-free rule predicts *whether* calibration will pay either — so the 200-tile cost buys the answer and the question together.
- The fitted vectors are constant across tiles, so they are right on average and can be wrong on any individual scene (Fig. 8, row 4).
- ConInfer's reproduction is imperfect — 36.99 against a published 39.33. We report both numbers.

Every negative in this project is reported with an oracle bound, so the reader learns where the headroom is rather than only that we failed to reach it.

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Summary

What was built

Not a new architecture. **Two lines of arithmetic** on top of an existing pipeline: a per-class threshold vector and a per-class scale before the `argmax`, fitted on ∼200 labelled tiles with **no gradient steps**.

- +2.32 LoveDA   +5.67 Potsdam   +2.51 on ConInfer's CLIP pipeline
- Verified end-to-end by the official evaluator on held-out tiles
- A proof that per-class τ is the complete family given the argmax
- A causal experiment with pre-registered predictions
- 12 label-free attempts, bounded, with a stated mechanism

Target venue: IEEE Transactions on Geoscience and Remote Sensing

Thank You